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ORIGINAL ARTICLE

Absolute lung cancer risk increases among individuals with >15 quit-years: Analyses to inform the update of the American Cancer Society lung cancer screening guidelines

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Abstract

Background: This report quantifies counteracting effects of quit-years and concomitant aging on lung cancer risk, especially on exceeding 15 quit-years, when the US Preventive Services Task Force (USPSTF) recommends curtailing lung-cancer screening.

Methods: Cox models were fitted to estimate absolute lung cancer risk among Prostate, Lung, Colorectal and Ovarian Cancer Screening Trial (PLCO) and National Lung Screening Trial (NLST) participants who ever smoked. Absolute lung cancer risk and gainable years of life from screening for individuals aged 50 to 80 in the US-representative National Health Interview Survey (NHIS) 2015–2018 who ever smoked were projected. Relaxing USPSTF recommendations to 20/25/30 quit-years versus augmenting USPSTF criteria with individuals whose estimated gain in life expectancy from screening exceeded 16.2 days according to the Life Years From Screening-CT (LYFS-CT) prediction model was compared.

Results: Absolute lung cancer risk increased by 8.7%/year (95% CI, 7.7%–9.7%; $p < .001$) as individuals aged beyond 15 quit-years in the PLCO, with similar results in NHIS and NLST. For example, mean 5-year lung cancer risk for those aged 65 years with 15 quit-years = 1.47% (95% CI, 1.35%–1.59%) versus 1.76% (95% CI, 1.62%–1.90%) for those aged 70 years with 20 quit-years in the PLCO. Removing the quit-year criterion would make 4.9 million more people eligible and increase the proportion of preventable lung cancer deaths prevented (sensitivity) from 63.7% to 74.2%. Alternatively, augmentation using LYFS-CT would make 1.7 million more people eligible while increasing the lung cancer death sensitivity to 74.0%.

Conclusions: Because of aging, absolute lung cancer risk increases beyond 15 quit-years, which does not support exemption from screening or curtailing screening once it has been initiated. Compared with relaxing the USPSTF quit-year criterion,

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augmentation using LYFS-CT could prevent most of the deaths at substantially superior efficiency, while also preventing deaths among individuals who currently smoke with low intensity or long duration.

KEYWORDS
lung cancer, precision prevention, quit, risk, screening, smoking, USPSTF

BACKGROUND

It is well-known that the relative risk of lung cancer decreases over time in individuals who formerly smoked versus individuals who continue smoking.[1–3] However, it is less well-known that the absolute risk of lung cancer does not concomitantly decrease over time in individuals who formerly smoked; as they accumulate quit-years, they also age, and these effects counteract each other on the absolute risk scale. Previous research suggests that absolute lung cancer risk does not decrease as individuals who formerly smoked age.[4,5] Because absolute risk of lung cancer death is the primary determinant of the benefit of lung cancer screening for an individual,[6] the benefit of screening for individuals who formerly smoked may not decrease over time, as is commonly presumed.

In 2021, the US Preventive Services Task Force (USPSTF) expanded lung cancer screening with low-dose computed tomography (CT) to adults aged 50 to 80 years with ≥20 pack-years, but continued to restrict screening for individuals who formerly smoked to ≤15 quit-years.[7] However, if absolute lung cancer risk increases as individuals with >15 quit-years grow older, these individuals could increasingly benefit from continued screening. The CISNET consortium projected that extending USPSTF criteria to include those with ≤25 quit-years was on the efficiency frontier for both deaths prevented and life-years gained.[8] Furthermore, augmenting USPSTF criteria to include individuals aged 50 to 80 years who are missed by current USPSTF criteria, who would gain the most days of life from attending lung cancer screening according to prediction models, may increase the number of preventable deaths and gainable life-years from screening, and may reduce racial/ethnic health disparities.[9,10] For this purpose, American College of Chest Physician guidelines recommend using the Life Years From Screening-CT (LYFS-CT) prediction-model of life-years gained from screening to inform screening eligibility.[11]

We examine two separate, but related, issues. First, we quantified the counteracting effects of quit-years and concomitant aging on lung cancer risk in the Prostate, Lung, Colorectal and Ovarian (PLCO) Cancer Screening Trial, the National Lung Screening Trial (NLST), and the National Health Interview Survey (NHIS), focusing on risk in individuals with >15 quit-years. This is purely an epidemiologic analysis. Second, we used the NHIS to project the number of preventable lung cancer deaths and life-years gainable in the United States from attending screening for a range of quit-year criteria, and by augmenting USPSTF criteria with individuals who gain the most life-years from screening as chosen by LYFS-CT.[9]

METHODS

We fit Cox models to estimate 5-year absolute risk of lung cancer in individuals aged 55 to 74 years who smoked ≥20 pack-years in two cohorts: PLCO,[12] which recruited US individuals from 1993 to 2001, and the chest radiography arm of the NLST, which recruited US individuals with ≥30 pack-years from 2002 to 2004.[13] The risk models use standard lung-cancer risk factors, and allowed for competing mortality; the hazard ratios for the lung cancer incidence and competing mortality models when fit to each of the PLCO and NLST are in Table S1. We additionally considered using age and quit-years as categorical variables in the PLCO; these results are shown in the Supplement.

We define the age at which an individual quit smoking (quit-age) as an individual's current age minus the number of years since they quit smoking (quit-years) and define individuals who currently smoke to have 0 quit-years. Stratified by quit-age, we then plotted estimated 5-year lung cancer risk (allowing for competing mortality) as quit-years (and age) increase each year for both PLCO and NLST. Individuals with the same quit-age but different numbers of quit-years are different people, based on their characteristics at recruitment to the study.

For the PLCO, we additionally plotted 5-year lung cancer risks starting with individuals who had at least 15 quit-years along a lexis diagram of age and quit-years. For example, we first estimated 5-year lung cancer risk among those aged 55 years with 15 quit-years at study enrollment; we next estimated risk among those aged 56 years with 16 quit-years at enrollment, then estimated risk among those aged 57 years with 17 quit-years at enrollment, and so on. Different individuals are included at each point along a quit-age line. It was not possible to make a similar plot in NLST because individuals with >15 quit-years at enrollment were ineligible for the NLST.

Linear regression was used to calculate the annual percentage change (APC) in risk relative to individuals who still smoked with the same quit-age. The model regressed current age on the APC in risk relative to individuals who still smoked (i.e., individuals with 0 quit-years) who would have the same quit-age, for quit-ages 55, 60, 65, and 70 years. From this, we assessed the APC over the initial decrease in risk and return to baseline risk. For the PLCO and NLST, we additionally calculated the APC relative to the risk when the risk returned to baseline levels for individuals with the same quit-age.

Using the 2015–2018 NHIS,[14] a US-representative sample, we calculated 5-year lung cancer risk based on the validated Lung Cancer Risk Assessment Tool (LCRAT) lung cancer incidence model[15]

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for all individuals aged 50 to 80 years with ≥20 pack-years. Details on the NHIS, and the multiple imputation for those with missing risk factor information, are in the Supplement. The LCRAT is a model of 5-year absolute risk of lung cancer, allowing for competing mortality. For each individual, we also calculated their individual life-years gainable from lung cancer screening based on the validated LYFS-CT model[9] (Table S2). LYFS-CT models the life-years gained from attending an NLST-like (three annual low-dose CT screens) lung screening program, by modeling life expectancy in the absence of screening, then reducing the risk of lung cancer death by 20.4% for 5 years, as observed in the NLST. More details on these models are provided in the Supplement.

We also used the 2015–2018 NHIS to estimate the number of individuals aged 50 to 80 years with ≥20 pack-years who are ineligible under USPSTF 2021 recommendations because of having quit smoking ≥16 years earlier. We additionally estimated the number who would be eligible if the quit-year criterion was increased to 20, 25, or 30 years. Among these individuals, we projected (using previously published methodology[15]) the sensitivity of preventable lung cancer deaths (the proportion of preventable lung cancer deaths prevented by screening those eligible for screening), estimated using the Lung Cancer Death Risk Assessment Tool model,[15] sensitivity of gainable years of life, estimated using the LYFS-CT model,[9] and screening efficiency (number needed to screen [NNS] per lung cancer death prevented and per 10 life-years gained). For the additional individuals who would be eligible under each of the scenarios considered, results were compared with the same metrics calculated among individuals who became eligible under USPSTF 2021 recommendations but would not have been eligible under USPSTF 2013 recommendations (aged 55–80 years, ≥30 pack-years, no more than 15 quit-years). To evaluate the benefit of augmenting USPSTF criteria with individuals who were predicted to gain the most life-years from screening, we identified individuals predicted by LYFS-CT to gain ≥16.2 days of life, who are eligible under American College of Chest Physician guidelines[11] but ineligible under USPSTF 2021. The threshold of 16.2 days selected the same number of individuals for screening as USPSTF 2013 guidelines.[9] We also considered augmenting USPSTF 2021 with individuals eligible under a lower threshold of 12 days of life, which has been used previously,[10] and augmenting USPSTF 2021 with no quit-year criteria with individuals predicted by LYFS-CT to gain ≥16.2 days of life. Finally, we evaluated these potential eligibility criteria by self-reported race/ethnicity. Analyses were performed in R v4.0.2, using the lcmodels package.[16]

RESULTS

The characteristics of individuals who ever-smoked (≥20 pack-years) in the PLCO, the chest radiography arm of NLST (where all participants had ≥30 pack-years), and NHIS 2015-2018, the three data sets analyzed in this study, are shown in Table S3, demonstrating the differences between trial participants and population samples, and changes over time in the characteristics of individuals who ever smoked. In a contemporary, population-representative sample of individuals aged 50 to 80 who have ever smoked, half of US individuals who formerly smoked in the NHIS (50.5%) had ≥15 quit-years (Table S3).

Figure 1 shows the 5-year lung cancer risk in the PLCO, restricted to individuals with ≥20 pack-years, along a lexis diagram of age and quit-years, stratified by the age at quitting (Figure 1A). This approach stitched together the observed risks among people as age and quit-years each advance by 1 year, starting at the age of quitting smoking. For all the figures, the lines connect the risks between individuals with the same quit-age, but each point on the line shows the risk among different individuals, based on their characteristics when they entered the study.

Initially, risk decreases for all quit-ages (relative annual percentage decrease [RAPC] in the first 5 quit-years: −4.4%, 95% CI, −6.1 to −2.9%; $p < .001$). However, as aging counteracts the effect of increasing quit-years, risk rebounded by 10 quit-years (RAPC over the first 10 quit-years: −0.3%; 95% CI, −1.3 to 0.6%, $p = .50$). After the first 10 quit-years, the effect of aging outweighed the effect of increasing quit-years, and thus risk increased (RAPC beyond 10 quit-years: 3.8%; 95% CI, 2.6–5.0; $p < .001$). Estimated risks by time since quitting in the NLST show the same general pattern (Figure S1, Supplementary Results), as do estimated risks in the PLCO when categorizing age and quit-years (Figure S2A), implying that the results are not limited to one form of parameterization.

Importantly, Figure 1B shows that risk substantially increases beyond 15 quit-years for individuals who entered the PLCO with ≥20 pack-years and 15 quit-years (RAPC beyond 15 quit-years: 8.7%; 95% CI, 7.7–9.7; $p < .001$). Similar results were seen when categorizing age and quit-years (Supplementary Results, Figure S2B). Such individuals become ineligible for screening under USPSTF recommendations, but their lung cancer risk continues to increase.

Figure 2 shows the mean 5-year lung cancer risk (on a log scale), as estimated by the LCRAT, among individuals with ≥20 pack-years in NHIS 2015–2018. For individuals with the same quit-age who quit smoking before aged 65 years, the risks increase for individuals who are older with more quit-years (RAPC, 3.3%; 95% CI, 2.4–4.2; $p < .001$). Table S4 shows how the characteristics of individuals who quit at age 50 years change with increasing age/quit-years. Mean pack-years increases with increasing age, which contributes to the increase in lung cancer risk, in addition to the increase in age.

Figure 3 shows the average days of life gained from screening (on a log scale), as estimated by the LYFS-CT model, among the same individuals as in Figure 2. For individuals who quit smoking at or before age 50 years, life-years gained from lung screening increases even as individuals age (RAPC, 1.8%; 95% CI, 0.89–2.77; $p = .013$), and life-years gained remained fairly constant for individuals who quit at age 55 years (range, 11.93–14.31 days at ages 55–80 years). For individuals who quit at age 60 years or older, the gainable life-years do not return to the level when they quit smoking but remain very high. Figures 2 and 3 demonstrate that the current population of individuals aged 50 to 70 years with ≥20 pack-years

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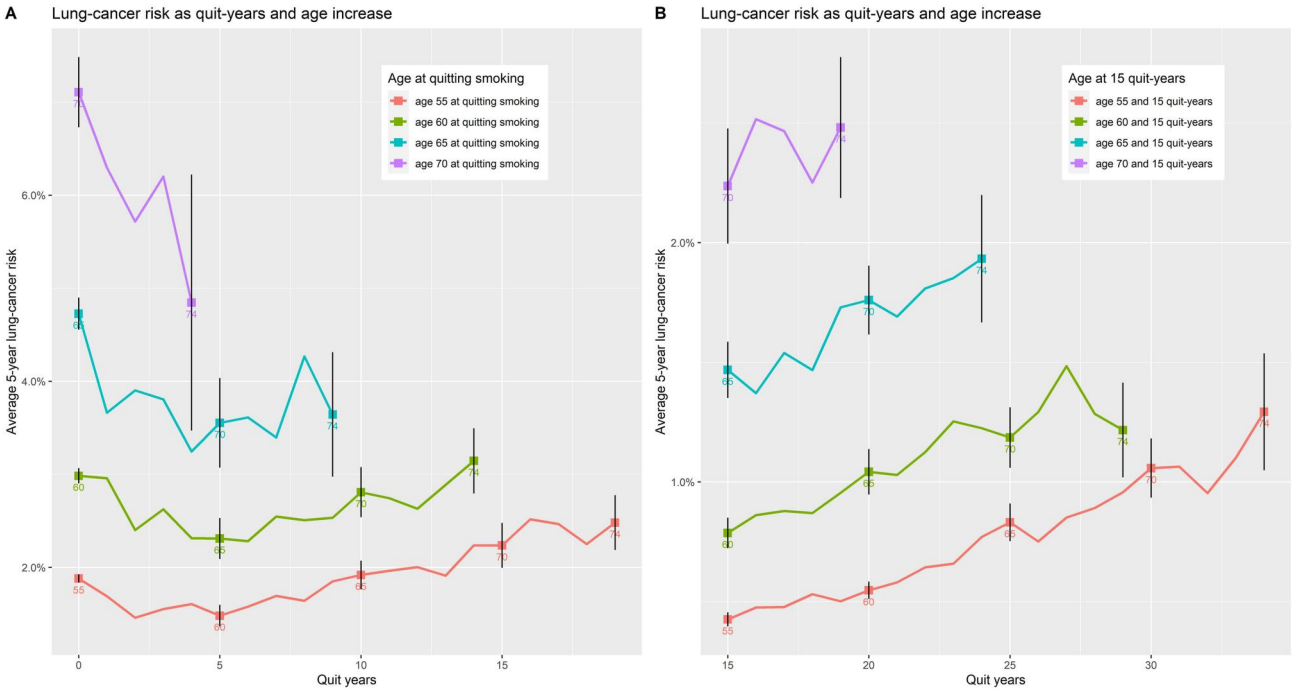


FIGURE 1 Five-year lung cancer risk and 95% CIs by quit-years among (A) ever-smokers who quit at ages 55, 60, 65, and 70 years in the PLCO and (B) ever-smokers with ≥15 quit-years who had 15 quit-years at ages 55, 60, 65, and 70 years in the PLCO. Relative annual percentage change in the first 5 quit-years: −4.4%; 95% CI, −6.1 to −2.9; $p < .001$ (A). Relative annual percentage change in the first 10 quit-years: −0.3%; 95% CI, −1.3 to 0.6; $p = .50$ (A). Relative annual percentage change beyond 15 quit-years: PLCO, 3.8%; 95% CI, 2.6–5.0; $p < .001$ (B). Number contributing to each line of data: (A) Quit-age 55: $N = 3348$. Quit-age 60: $N = 2424$. Quit-age 65: $N = =1157$. Quit-age 70: $N = 471$. (B) Age 55 and 15 quit-years: $N = 2193$. Age 60 and 15 quit-years: $N = 1352$. Age 65 and 15 quit-years: $N = 877$. Age 70 and 15 quit-years: $N = 287$. We note that individuals on the same quit-age line are different individuals at each point. PLCO indicates Prostate, Lung, Colorectal and Ovarian Cancer Screening Trial.

have higher lung cancer risk and higher life-gained from attending screening at ≥20 quit-years than individuals with 15 quit-years.

The impact of the quit-year criteria on screening eligibility of individuals who currently smoke with ≥20 pack-years

Using the NHIS 2015–2018, we estimate that 14.3 million people aged 50 to 80 years who ever smoked (32%) are eligible for screening under USPSTF 2021 recommendations. The characteristics of these individuals are shown in Table S3. If the quit-year criteria were increased to 20 quit-years, 25 quit-years, 30 quit-years, or removed, an additional 1.6 million, 2.5 million, 3.7 million, and 4.9 million individuals who ever-smoked would be eligible, respectively. If the quit-year criterion was removed, this would result in 43% of individuals aged 50 to 80 years who ever-smoked being eligible (Table 1), an 11% absolute increase or 34% relative increase. More than one-half of those who would become eligible are aged 70 to 80 years (52%), with 59% having quit smoking ≥25 years earlier and 13% having 16 to 19 quit-years (Table 2). Among US individuals who formerly smoked who are ineligible by USPSTF 2021 solely because of >15 quit-years, 56% have ≥30 pack-years, and 17% have ≥50 pack-years.

Increasing or removing the quit-year criterion could increase lung cancer death sensitivity from 63.7% to 67.4% (20 quit-years), 69.5% (25-quit-years), 71.9% (30 quit-years), or 74.2% (removing the quit-year criterion), corresponding to an additional 2914, 4597, 6485, and 8275 preventable lung cancer deaths over 5 years, respectively. Life-years gained sensitivity could increase from 60.5% to 64.0% (20 quit-years), 65.9% (25 quit-years), 68.0% (30 quit-years), and 70.0% (removing the quit-year criterion), corresponding to an additional 41,984, 65,175, 91,188, and 115,107 gainable life-years, respectively (Table 1).

Screening efficiency in the 4.9 million individuals with >15 quit-years who would become eligible if the quit-year criterion was increased/removed was comparable to the 6.2 million individuals newly eligible by USPSTF 2021 recommendations (Table 1). The NNS to prevent one lung cancer death, based on three annual screens and 5 years of follow-up (NLST-like screening), may be superior if the quit-year criterion was removed (NNS = 592 in ineligible individuals with >15 quit-years vs. NNS = 634 in the newly eligible), although the NNS to gain 10 life-years may be inferior (NNS = 426 in ineligible individuals with >15 quit-years vs. 346 in the newly eligible). The NNS to prevent one death appears superior because ineligible individuals with >15 quit-years are projected to have higher risk than the newly eligible. However, the NNS to gain 10 life-years appears inferior because individuals with >15 quit-years are older than the

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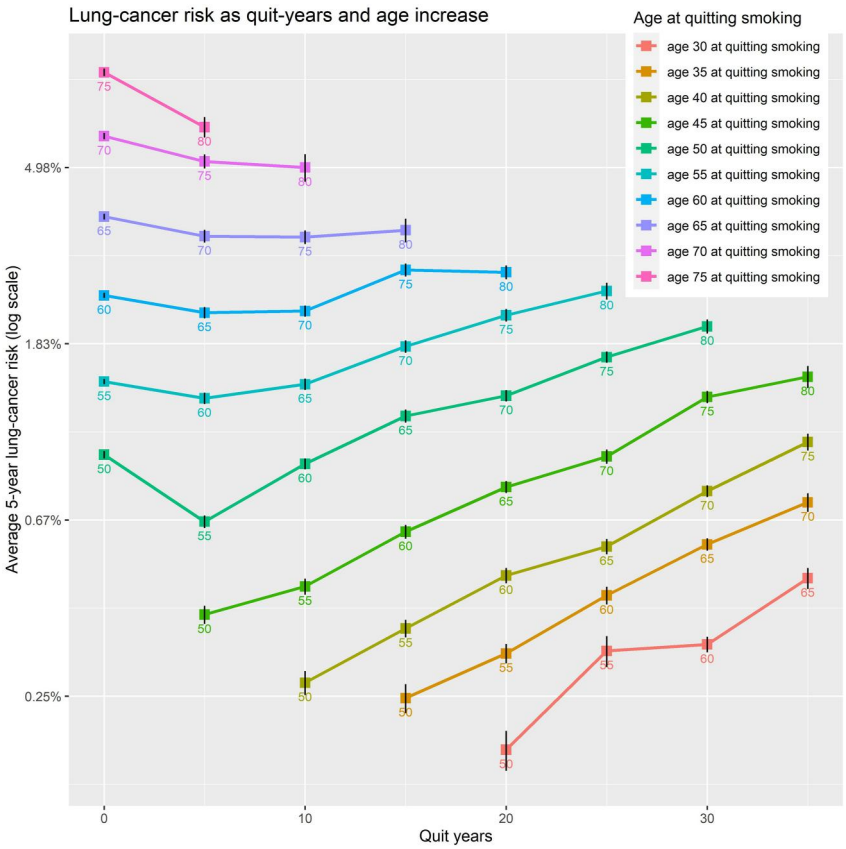


FIGURE 2 Five-year lung cancer risk and 95% CIs, shown on a log scale, by quit-age and quit-years among individuals with ≥20 pack-years in the 2015–2018 NHIS. 5-year lung cancer risks were calculated using the LCRAT model.[15] Relative annual percentage change 3.3%; 95% CI, 2.4–4.2; $p < .001$. We note that individuals on the same quit-age line are different individuals at each point. LCRAT indicates Lung Cancer Risk Assessment Tool; NHIS, National Health Interview Survey.

newly eligible (Table 2). When individuals who are ineligible under USPSTF 2021 are stratified by their quit-years, screening becomes less efficient with increasing quit-years (Table 1) but remains more efficient per lung cancer death prevented up to 30 quit-years.

Augmenting USPSTF 2021 to include individuals identified by LYFS-CT

Table 1 considers augmenting USPSTF 2021 to also include individuals estimated to gain ≥16.2 days of life from screening according to the LYFS-CT model. This augmentation results in one-third as many newly eligible people as by removing the quit-year criteria (1.7 vs. 4.9 M, respectively), yet could save a very similar number of lives (8137 vs. 8275, respectively) and gain 90% of the life-years (103,933 vs. 115,107, respectively). Hence, augmentation has substantially superior NNS for preventing death (NNS = 212 vs. 592, respectively) and gaining 10 life-years (NNS = 166 vs. 426, respectively) (Table 1). Augmentation has the additional feature of also including the individuals who are estimated to gain the most life-years from screening who currently smoke and are missed by USPSTF 2021. These NNS levels are much lower than for the people additionally eligible under USPSTF 2021.

Among individuals who formerly smoked, augmentation identifies 0.5 million individuals with >15 quit-years, and they have substantially superior NNS to prevent one lung cancer death (NNS = 210) and to gain 10 years of life (NNS = 174) than any incremental loosening of the quit-year criteria (Table 1). Notably, only one-third as many individuals with >15 quit-years would be additionally eligible under augmentation versus increasing the quit-year criterion to 20 years (1.6 million), but 84% of the lung cancer deaths could be prevented (2438 vs. 2914) and 70% of the life-years gained (29,483 vs. 41,984). Using a reduced threshold of ≥12 days of life gained for LYFS-CT would still screen fewer people, but now save substantially more lives and life-years, than removing the quit-year criterion (Table 1).

Because it may not be practical to calculate the benefit from screening under LYFS-CT for all individuals aged 50 to 80 year who ever-smoked and are ineligible under USPSTF 2021, Table 3 shows what proportion of individuals with given characteristics would become eligible under augmentation. Using a threshold of 16.2 days of life gained, 16% of USPSTF 2021–ineligible African Americans

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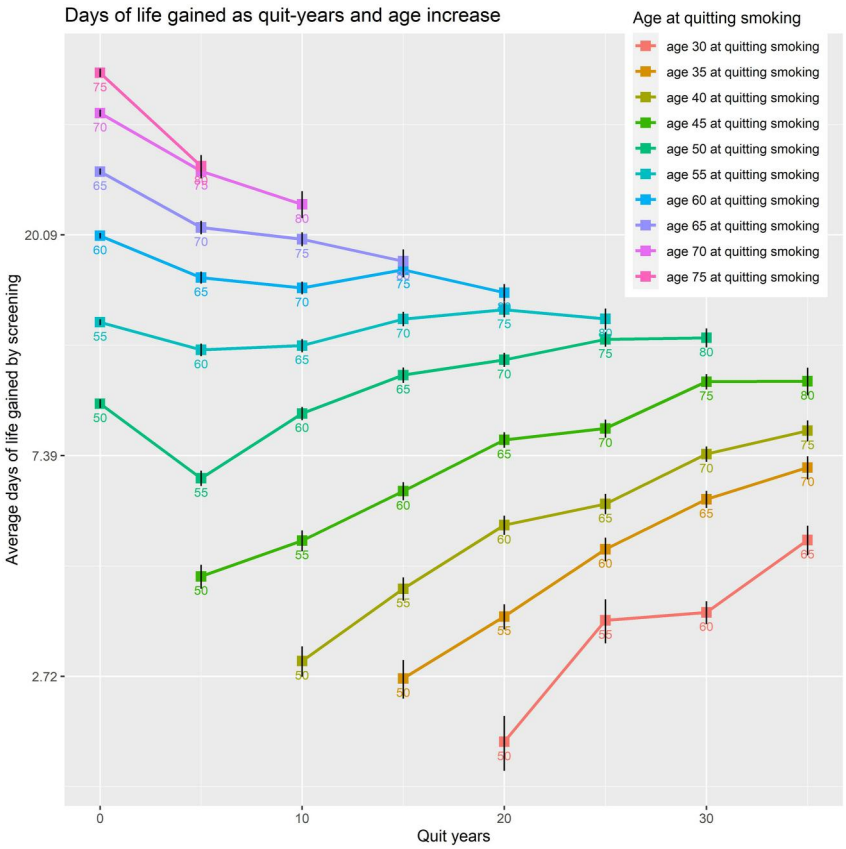


FIGURE 3 Days of life gained and 95% CIs, shown on a log scale, from screening by quit-years among individuals with ≥20 pack-years in the 2015–2018 NHIS. Days of life gained from screening were calculated using the LYFS-CT model.[9] Relative annual percentage change: 1.8%; 95% CI, 0.9–2.8%; $p = .013$. We note that individuals on the same quit-age line are different individuals at each point. Age was categorized, with age 50 years representing individuals aged 50–52 years, age 55 years representing individuals aged 53–57 years, up to age 80 years representing individuals aged 78–80 years. Quit-years were categorized similarly, with individuals with 3–7 quit-years represented by 5 quit-years, individuals with 8–12 quit-years represented by 10 quit-years etc. Zero quit-years represent current smokers. LYFS-CT indicates Life Years From Screening-CT; NHIS, National Health Interview Survey.

would become eligible, compared with 3% to 5% of individuals from other races/ethnicities. Eighteen percent of individuals who currently smoke would become eligible, as would 20% and 38% of individuals who used to smoke with 40 to 49 and ≥50 pack-years, respectively.

Impact on eligibility by self-reported race/ethnicity and sex

When the quit-year criterion was relaxed to 20 years, 17% of eligible individuals were racial/ethnic minorities, compared with 19% when a very similar number were eligible under augmentation. The difference is due to 28% of the additional individuals eligible under augmentation being African Americans. The sensitivity for each self-reported race/ethnicity to both lung cancer death and life-years gained is higher under augmentation, most notably for African Americans (sensitivity to lung cancer death: 57% under 20 quit-years vs. 75% under augmentation), eliminating the difference in sensitivity between Whites and African Americans (Table 4). The differences in sensitivity between Whites and both Asian Americans and Hispanic Americans remained the same (14%–15% and 17%–18%, respectively).

Relaxing the quit-year criterion to 20 years increased the difference between men and women in the proportion of individuals aged 50 to 80 years who-ever smoked who were eligible (from 4% to 6% more for men; Table S5), in addition to increasing the difference in sensitivity to lung cancer deaths (4% to 5%) and life-years gained (5% to 6%). These gaps widened as the quit-year criteria increased, up to a 10% difference in eligibility when the quit-year criterion was removed, with 8% and 9% differences in lung cancer death and life-years gained sensitivity, respectively. In comparison, augmenting USPSTF 2021 guidelines with individuals who gain ≥16.2 days reduced the eligibility gap and differences in sensitivity to 2% and 3% for lung cancer death and life-years gained sensitivity, respectively.

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TABLE 1 The modeled performance of lung cancer screening among those eligible by (1) USPSTF 2013 recommendations, (2) USPSTF 2021 recommendations, (3) USPSTF 2021 recommendations with no quit-year criteria, (4) additional individuals eligible under USPSTF 2021 with no quit year criteria, and (5) additional individuals eligible under USPSTF 2021 compared with USPSTF 2013; in NHIS 2015–2018.

	Number eligible for screening		LCD prevented over 5 years			LYG		
	N	%	*N*	Sensitivity[a] (%)	NNS/LCD	*N*	Sensitivity[b] (%)	NNS/10 LYG
USPSTF 2013	8,084,495	18	40,424	51	200	550,630	46	151
USPSTF 2021	14,292,401	32	50,217	64	285	729,866	60	201
USPSTF 2021 with ≤20 quit-years	15,854,178	35	53,131	67	298	771,850	64	210
USPSTF 2021 with ≤25 quit-years	16,814,140	38	54,814	70	307	795,041	66	216
USPSTF 2021 with ≤30 quit-years	18,002,805	40	56,702	72	317	821,053	68	224
USPSTF 2021 with no quit year criteria	19,192,031	43	58,492	74	328	844,973	70	232
USPSTF 2021 or ≥16.2 days life by LYFS-CT[c]	16,014,364	36	58,354	74	274	833,799	69	192
USPSTF 2021 or ≥12 days life by LYFS-CT[c]	17,627,756	39	62,434	79	282	895,280	74	197
USPSTF 2021 with no quit-year criteria or ≥16.2 days life by LYFS-CT[c]	20,404,782	46	64,209	81	318	919,606	76	222
Comparing the extra individuals eligible for screening								
USPSTF 2021 vs. USPSTF 2013	6,207,906	14	9,793	12	634	179,236	15	346
USPSTF 2021 w/no quit-year criteria vs. USPSTF 2021	4,899,630	11	8,275	10	592	115,107	10	426
USPSTF 2021 or LYFS-CT[c] (≥16.2 days) vs. USPSTF 2021	1,748,963	4	8,137	10	215	103,933	9	168
USPSTF 2021 or LYFS-CT[c] (≥12 days) vs. USPSTF 2021	3,335,355	7	12,217	15	273	165,414	14	202
USPSTF 2021 with no quit-year criteria or LYFS-CT[c] (≥16.2 days) vs. USPSTF 2021	6,112,381	14	13,992	18	437	189,740	16	322
Comparing extra-eligible populations by quit-year criteria								
LYFS-CT (≥16.2 days) with >15 quit-years[c]	512,603	1	2438	3	210	29,483	2	174
LYFS-CT (≥12 days) with >15 quit-years[c]	1,177,893	3	4392	6	268	54,797	5	215
16-20 quit-years	1,561,777	3	2914	4	536	41,984	3	372
21-25 quit-years	959,962	2	1683	2	570	23,191	2	414
26-30 quit-years	1,188,665	3	1888	2	630	26,012	2	457
≥31 quit-years	1,189,226	3	1790	2	664	23,919	2	497

Note: These values are estimated from NHIS 2015–2018.

Abbreviations: LCD, lung cancer deaths; LYG, life-years gained; LYFS-CT, Life-Years From Screening-CT; NNS, number needed to screen; USPSTF, US Preventive Services Task Force.

[a]Proportion of preventable lung cancer deaths over 5 years prevented by screening eligible individuals compared with screening all ever-smokers aged 50–80 years.

[b]Proportion of gainable life-years gained by screening eligible individuals compared to screening all ever-smokers aged 50–80 years.

[c]Individuals who would gain ≥16.2 or ≥12 days of life by the LYFS-CT model.

DISCUSSION

New USPSTF lung cancer screening recommendations continue to curtail screening for individuals aged 50 to 80 years who exceed 15 quit-years, and the Centers for Medicare & Medicaid Services national coverage determination extended the age range to those aged 50 to 77 years and reduced the pack-year limit to 20 but retained the 15 quit-years limit.[17] However, we demonstrated that lung cancer risk increases beyond 5 quit-years in two cohorts (PLCO and NLST), and especially beyond 15 quit-years (PLCO) because the effect of aging outweighs the effect of increasing quit-years. The modeled gainable years of life from screening also increased beyond 15 quit-years, especially for those who quit before age 60 years. These findings do not support excluding individuals with >15 quit-years from screening at age 50 years, or curtailing screening in people aged >50 years when they achieve 16 quit-years.

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TABLE 2 Characteristics of ever-smokers from the NHIS 2015–2018: (1) eligible under USPSTF 2013 recommendations, (2) eligible under USPSTF 2021 but not USPSTF 2013, (3) everyone eligible under USPSTF 2021, (4) those ineligible under USPSTF 2021 who would be eligible without the quit-year criteria, and (5) those ineligible under USPSTF 2021 who would be eligible under augmentation with individuals who were predicted to gain ≥16.2 days or ≥12 days life by the LYFS-CT model.

	NHIS 2015–2018: eligible under USPSTF 2013		NHIS 2015–2018: additionally eligible under USPSTF 2021		NHIS 2015–2018: everyone eligible under USPSTF 2021		NHIS 2015–2018: additionally eligible beyond USPSTF 2021 if there was no quit-year criterion		NHIS 2015–2018: additionally eligible beyond USPSTF 2021 if augmenting with ≥ 16.2 days life by LYFS-CT		NHIS 2015–2018: additionally eligible beyond USPSTF 2021 if augmenting with ≥12 days life by LYFS-CT	
	N	%	*N*	%	*N*	%	*N*	%	*N*	%	*N*	%
Age (years)												
50–54	-	0	2,773,604	45	2,773,604	19	98,150	2	10,148	1	43,716	1
55–59	2,140,252	26	1,089,647	18	3,229,899	23	403,693	8	53,217	3	240,989	7
60–64	2,252,272	28	914,493	15	3,166,765	22	749,266	15	231,942	13	580,754	17
65–69	1,700,001	21	792,999	13	2,493,000	17	1,091,317	22	394,479	23	756,591	23
70–74	1,229,315	15	438,492	7	1,667,807	12	1,267,565	26	524,478	30	874,403	26
75–80	762,655	9	198,672	3	961,327	7	1,289,639	26	507,700	29	838,902	25
Self-reported race/ethnicity												
White	6,814,944	84	4,949,350	80	11,764,293	82	4,131,695	84	1,130,609	66	2,316,809	69
African American	615,483	8	606,859	10	1,222,341	9	333,937	7	478,446	28	788,946	24
Hispanic American	453,465	6	437,512	7	890,977	6	305,321	6	77,499	5	157,172	5
Asian American/other	200,604	2	214,186	3	414,790	3	128,676	3	35,409	2	72,427	2
Smoking status												
Current	4,194,695	52	4,094,090	66	8,288,785	58	-	0	1,090,402	63	1,823,909	55
Former	3,889,801	48	2,113,816	34	6,003,617	42	4,899,630	100	631,561	37	1,511,446	45
Sex												
Male	4,904,849	61	3,301,662	53	8,206,511	57	3,225,900	66	914,821	53	1,839,101	55
Female	3,179,646	39	2,906,244	47	6,085,890	43	1,673,729	34	807,142	47	1,496,254	45
Pack-years (current smokers)												
<20	-	0	-	0	-	0	-	0	1,090,402	100	1,823,909	100
20–29	-	0	2,714,273	66	2,714,273	33	-	0	-	0	-	0
30–39	1,149,984	27	896,597	22	2,046,581	25	-	0	-	0	-	0
40–49	1,498,769	36	229,997	6	1,728,766	21	-	0	-	0	-	0
≥50	1,545,941	37	253,223	6	1,799,164	22	-	0	-	0	-	0
Pack-years (former smokers)												
<20	-	0	-	0	-	0	-	0	122,349	19	358,040	24
20–29	-	0	1,617,048	76	1,617,048	27	2,175,517	44	496	0	4,537	0
30–39	1,178,265	30	282,065	13	1,460,330	24	1,284,853	26	65,221	10	274,917	18
40–49	1,042,512	27	76,923	4	1,119,435	19	593,904	12	119,992	19	301,977	20
≥50	1,669,024	43	137,780	7	1,806,804	30	845,355	17	323,503	51	571,975	38
Quit-years (former smokers)												
<5 years	1,404,182	36	820,252	39	2,224,434	37	-	0	96,564	15	220,034	15
5–9 years	1,052,639	27	556,951	26	1,609,590	27	-	0	1,544	0	36,460	2

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TABLE 2 (Continued)

	NHIS 2015–2018: eligible under USPSTF 2013		NHIS 2015–2018: additionally eligible under USPSTF 2021		NHIS 2015–2018: everyone eligible under USPSTF 2021		NHIS 2015–2018: additionally eligible beyond USPSTF 2021 if there was no quit-year criterion		NHIS 2015–2018: additionally eligible beyond USPSTF 2021 if augmenting with ≥ 16.2 days life by LYFS-CT		NHIS 2015–2018: additionally eligible beyond USPSTF 2021 if augmenting with ≥12 days life by LYFS-CT	
	N	%	*N*	%	*N*	%	*N*	%	*N*	%	*N*	%
10–14 years	1,027,963	26	537,586	25	1,565,549	26	-	0	107,594	17	226,372	15
15–19 years	405,016	10	199,028	9	604,044	10	640,713	13	198,928	31	398,788	26
20–24 years	-	0	-	0	-	0	1,357,525	28	207,776	33	566,028	37
≥25 years	-	0	-	0	-	0	2,901,393	59	19,155	3	63,765	4

Abbreviations: LCD, lung cancer deaths; LYG, life-years gained; LYFS-CT, Life-Years From Screening-CT; NNS, number needed to screen; USPSTF, US Preventive Services Task Force.

Furthermore, more than one-half (56%) of individuals aged 50 to 80 years with ≥20 pack-years and >15 quit-years who are not eligible under USPSTF 2021 have smoked ≥30 pack-years and 17% have smoked ≥50 pack-years. Consequently, we considered extending the quit-year criterion to the 11% of US individuals who formerly smoked who are ineligible solely because of >15 quit-years. Removing the quit-year criterion, we projected 10% increases in both preventable lung cancer deaths (8275 over 5 years) and gainable life-years (115,107), at an efficiency (NNS to prevent 1 death and to gain 10 life-years) comparable to those newly eligible under USPSTF 2021. Incrementally loosening the quit-year criterion to ≥20/25/30 provides incremental gains.

However, any quit-year criterion results in unintended outcomes. For example, all individuals with 20 quit-years are ineligible, yet some of these individuals would have been eligible 5 years ago, despite having more benefit from screening when they became ineligible than 5 years earlier. Such unintended outcomes can be overcome by removing the quit-year criterion, but this widens both the male-female and White-African American gaps in eligibility, lives saved, and life-years gained. These difficulties are naturally overcome by augmenting eligibility criteria to also include individuals estimated to gain ≥16.2 days of life by a prediction model such as LYFS-CT, as now recommended by the American College of Chest Physicians,[11] which is available as a clinical decision support tool accessible via the Web[18] and in a form that is integrated with the electronic health record.[19] Augmenting USPSTF 2021 with people with ≥16.2 days of life gained from screening (according to LYFS-CT) would screen only one-third as many individuals with >15 quit-years as if the quit-year criterion was increased to 20 quit-years, yet could prevent 84% of the lung cancer deaths and gain 70% of life-years. Compared with removing the quit-year criterion, augmenting would screen only 10% of people with >15 quit-years, yet possibly prevent 29% of the lung cancer deaths and gain 26% of the life-years. Additionally, augmenting reduced both the male-female and White-African American gaps in eligibility, lives saved, and life-years gained, which the quit-year criterion are projected to widen.

Our findings broadly agree with CISNET modeling, which suggests that criteria including individuals with up to 25 quit-years dominates 2021 USPSTF recommendations and are on the efficiency frontier for both deaths prevented and life-years gained.[8] The older ages of those who are ineligible because of the quit-year criteria lower their gainable life-years but increases the number of preventable deaths. Information on the benefit of attending screening could inform a shared decision-making session, along with information on the risks of potential harms, which may be preferable to an upper age limit for screening.

Most previous studies which have investigated lung cancer risk/lung cancer death risk among individuals who formerly smoked provide relative risks, relative to individuals who currently smoke.[1,3,20] In 2004, IARC[21] concluded that "stopping smoking at any age avoids the further increase in risk of lung cancer incurred by continued smoking," and that “the younger the age at cessation, the greater the benefit."[2,22] Our absolute risk analysis also demonstrates that, the younger the age at quitting smoking, the lower the future absolute risk of lung cancer. These studies make clear the importance of quitting smoking as soon as possible.

In February 2022, US President Joe Biden set the ambitious goal of reducing the death rate from cancer by at least 50% over the next 25 years.[23] Increasing access to, and uptake of, lung cancer screening (with concomitant access to smoking-cessation services[25]) may be vital to ensure a population-level impact on cancer mortality.[23] Removing the quit-year criteria simplifies the identification of screening eligible individuals,[24] but includes people with lower gain in life expectancy from screening, may widen male-female and White–African American disparities in eligibility, and individuals who formerly smoked are not candidates for smoking-cessation services. In contrast, augmenting current criteria with those chosen by prediction models may concentrate a similar number of lives saved/life-years gained in fewer people eligible, may reduce male-female and White–African American disparities, and increases opportunities for smoking cessation by including individuals who currently smoke. Implementing prediction-based

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TABLE 3 Percentage of individuals aged 50–80 years who ever-smoked but are ineligible for screening under USPSTF that would be eligible under LYFS-CT using a threshold of 16.2 or 12 days of life-gained.

	NHIS 2015–2018: individuals aged 50–80 years who ever-smoked		NHIS 2015–2018: eligible under USPSTF 2021		NHIS 2015–2018: ineligible by USPSTF 2021 but ≥16.2 days life by LYFS-CT		NHIS 2015–2018: ineligible by USPSTF 2021 but ≥ 12 days life by LYFS-CT	
	N	Column %	*N*	% of individuals aged 50–80 years who ever-smoked	*N*	% of individuals aged 50–80 years who ever-smoked but are ineligible under USPSTF 2021	*N*	% of individuals aged 50–80 years who ever-smoked but are ineligible under USPSTF 2021
Age (years)								
50–54	8,352,815	19	2,773,604	33	10,148	0	43,716	1
55–59	9,018,530	20	3,229,899	36	53,217	1	240,989	4
60–64	8,885,275	20	3,166,765	36	231,942	4	580,754	10
65–69	7,667,347	17	2,493,000	33	394,479	8	756,591	15
70–74	6,159,314	14	1,667,807	27	524,478	12	874,403	19
75–80	4,669,931	10	961,327	21	507,700	14	838,902	23
Self-reported race/ethnicity								
White	34,973,814	78	11,764,293	34	1,130,609	5	2,316,809	10
African American	4,301,164	10	1,222,341	28	478,446	16	788,946	26
Hispanic American	3,866,398	9	890,977	23	77,499	3	157,172	5
Asian American/other	1,611,836	4	414,790	26	35,409	3	72,427	6
Smoking status								
Current	14,320,008	32	8,288,785	58	1,090,402	18	1,823,909	30
Former	30,433,205	68	6,003,617	20	631,561	3	1,511,446	6
Sex								
Male	24,157,628	54	8,206,511	34	914,821	6	1,839,101	12
Female	20,595,584	46	6,085,890	30	807,142	6	1,496,254	10
Pack-years (individuals who currently smoke)								
<20	6,031,223	13	-	0	1,090,402	18	1,823,909	30
20–29	2,714,273	6	2,714,273	100	-	0	-	0
30–39	2,046,581	5	2,046,581	100	-	0	-	0
40–49	1,728,766	4	1,728,766	100	-	0	-	0
≥50	1,799,164	4	1,799,164	100	-	0	-	0
Pack-years (individuals who formerly smoked)								
<20	19,529,958	44	-	0	122,349	1	358,040	2
20–29	3,792,565	8	1,617,048	43	496	0	4537	0
30–39	2,745,183	6	1,460,330	53	65,221	5	274,917	21
40–49	1,713,339	4	1,119,435	65	119,992	20	301,977	51
≥50	2,652,159	6	1,806,804	68	323,503	38	571,975	68
Quit-years (individuals who formerly smoked)								
<5 years	3,664,799	8	2,224,434	61	96,564	7	220,034	15
5–9 years	2,984,374	7	1,609,590	54	1544	0	36,460	3

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TABLE 3 (Continued)

	NHIS 2015–2018: individuals aged 50–80 years who ever-smoked		NHIS 2015–2018: eligible under USPSTF 2021		NHIS 2015–2018: ineligible by USPSTF 2021 but ≥16.2 days life by LYFS-CT		NHIS 2015–2018: ineligible by USPSTF 2021 but ≥ 12 days life by LYFS-CT	
	N	Column %	*N*	% of individuals aged 50–80 years who ever-smoked	*N*	% of individuals aged 50–80 years who ever-smoked but are ineligible under USPSTF 2021	*N*	% of individuals aged 50–80 years who ever-smoked but are ineligible under USPSTF 2021
10–14 years	2,714,374	6	1,565,549	58	107,594	9	226,372	20
15–19 years	3,465,603	8	604,044	17	198,928	7	398,788	14
20–24 years	14,879,006	33	-	0	207,776	1	566,028	4
≥25 years	2,725,048	6	-	0	19,155	1	63,765	2

Abbreviations: LYFS-CT, Life-Years From Screening-CT; NHIS, National Health Interview Survey; USPSTF, US Preventive Services Task Force.

screening eligibility so that it is practical even in low-income settings is a research priority.

A strength of our study is that we have shown consistent results in three cohorts with very different smoking characteristics. The different characteristics of individuals in the PLCO, NLST control arm, and NHIS reflect the eligibility criteria for the studies, how participants in trials differ from the general population, and how the characteristics of individuals who smoke have changed over time. The PLCO recruited individuals between 1993 and 2001, with NLST participants recruited between 2002 and 2004; the results in these cohorts therefore represent individuals who smoke from a wider range of birth cohorts along the same "quit-age" line, whereas in the NHIS, we evaluated the risks and gainable life-years among the individuals currently impacted by the USPSTF screening recommendations.

It is important to note that there are cohort effects in these estimates, and the same risks may not be observed in future individuals with the same attained age and age at quitting. For example, Table S4 shows the characteristics of individuals every 5 quit-years for individuals who quit at ages 50 and 65 years. For those aged 70 to 80 years, individuals who had 5 to 15 quit-years had more pack-years of smoking exposure compared with individuals the same age with 15 more quit-years. This would imply that future cohorts of individuals who formerly smoked may not have the same lung cancer risk as individuals who formerly smoked in 2022 because of differences in their smoking habits (as well as other risk factors) across their lives. Because both the characteristics of individuals who ever-smoked[26,27] and the contents of the cigarettes[28,29] change over time, we show that the general results seen in the PLCO and NLST, which do not represent individuals who ever-smoked in the general population in 2022, even when restricted to the same age and pack-year eligibility criteria, agree with the results in general population studies.

Our study has limitations. Our analyses assume that the mortality reduction from screening in the United States is the same as in the NLST. This assumption is supported by the lack of statistical heterogeneity in the relative reduction in lung cancer mortality across risk quintiles for low-dose CT versus chest radiography in the NLST,[30] although results have varied between studies with different screening and follow-up protocols,[31,32] and treatment advances will also change the mortality benefit of screening. Uptake of lung cancer screening is low,[33] and there is low adherence to screening guidelines,[34] both of which limit the population-level impact of screening. Some individuals quit smoking because of the onset of symptoms or diagnosis of serious disease.[35] Individuals in the PLCO or NLST who recently quit smoking did not die from a serious illness shortly after quitting and may therefore not be representative of individuals who recently quit smoking. We have not modeled the harms of screening, such as the number of false-positive screens, which may result in anxiety, and less frequently, unnecessary invasive procedures[7]; however, we note that extended follow-up of the NLST demonstrated that lung cancer screening resulted in little, if any, overdiagnosis.[36] Additional cost-effectiveness analyses are necessary to consider removing the quit-year criterion completely.

Individuals with >15 quit-years are not the only people who may be excluded from lung screening under USPSTF 2021 recommendations despite having higher risk and benefit than some eligible individuals; any guidelines based on smoking history and age cut-points will result in suboptimal screening recommendations. For example, previous research has shown that African Americans who are estimated to gain ≥16.2 days of life continue to be disproportionately ineligible by 2021 USPSTF recommendations.[10] However, risk- or benefit-based guidelines could ameliorate these situations. Everyone estimated to gain a certain number of days of life from screening, regardless of smoking history, race/ethnicity, sex, or other factors, could be identified by individualized prediction models for life-years gained from screening, such as LYFS-CT,

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TABLE 4 The modeled performance of lung cancer screening eligibility criteria, by self-reported race/ethnicity, in NHIS 2015–2018.

	Number eligible for screening			LCD prevented over 5 years				LYG			
	N	%	Absolute difference in % eligible vs. Whites	*N*	Sensitivity (%)	Absolute difference in sensitivity vs. Whites	NNS/LCD	*N*	Sensitivity (%)	Absolute difference in sensitivity vs. Whites	NNS/10 LYG
USPSTF 2021											
White	11,764,293	34		41,903	67		281	614,138	64		192
African American	1,222,341	28	5	5760	54	13	212	73,546	49	15	166
Asian American	316,316	25	9	655	52	14	483	11,583	49	15	273
Hispanic American	890,977	23	11	1728	48	18	516	27,373	44	20	325
USPSTF 2021 with ≤20 quit-years											
White	13,040,612	37		44,298	70		294	648,436	67		201
African American	1,361,440	32	6	6105	57	14	223	78,462	52	15	174
Asian American	348,244	27	10	685	55	16	508	12,138	51	16	287
Hispanic American	994,272	26	12	1858	52	19	535	29,346	47	20	339
USPSTF 2021 with ≤25 quit-years											
White	13,832,843	40		45,675	73		303	667,579	69		207
African American	1,437,629	33	6%	6303	59	14	228	80,827	53	16	178
Asian American	370,601	29	11%	715	57	15	518	12,543	53	16	295
Hispanic American	1,058,969	27	12%	1933	54	19	548	30,543	49	20	347
USPSTF 2021 with ≤30 quit-years											
White	14,836,613	42		47,267	75		314	689,540	72		215
African American	1,506,882	35	7	6477	60	15	233	82,910	55	17	182
Asian American	402,651	32	11	753	60	15	535	13,159	56	16	306
Hispanic American	1,139,342	29	13	2011	56	19	566	31,829	51	20	358
USPSTF 2021 with no quit year criteria											
White	15,895,989	45		48,869	78		325	710,941	74		224
African American	1,556,278	36	9%	6578	61	16	237	84,158	56	18	185
Asian American	424,833	33	12	773	62	16	550	13,502	57	17	315
Hispanic American	1,196,298	31	15	2,078	58	20	576	32,738	53	21	365

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TABLE 4 (Continued)

	Number eligible for screening			LCD prevented over 5 years				LYG			
	N	%	Absolute difference in % eligible vs. Whites	*N*	Sensitivity (%)	Absolute difference in sensitivity vs. Whites	NNS/LCD	*N*	Sensitivity (%)	Absolute difference in sensitivity vs. Whites	NNS/10 LYG
USPSTF 2021 or ≥16.2 days life by LYFS-CT[a]											
White	12,894,902	37		47,233	75		273	681,507	71		189
African American	1,700,788	40	−3	8097	75	0	210	103,894	69	2	164
Asian American	347,335	27	10	765	61	14	454	13,254	56	15	262
Hispanic American	968,476	25	12	2070	58	17	468	31,690	51	20	306
USPSTF 2021 or ≥12 days life by LYFS-CT[a]											
White	14,081,103	40		50,302	80		280	726,795	75		194
African American	2,011,287	47	−6	8826	82	−2%	228	115,702	77	−1	174
Asian American	377,551	30	11	835	67	13%	452	14,398	61	15	262
Hispanic American	1,048,149	27	13	2269	63	17	462	34,672	56	20	302
USPSTF 2021 with no quit year criteria or ≥16.2 days life by LYFS-CT[a]											
White	16,611,125	47		52,244	83		318	754,769	78		220
African American	1,953,666	45	2	8500	79	4	230	109,427	72	6	179
Asian American	453,527	36	12	875	70	13	519	15,052	64	15	301
Hispanic American	1,263,493	33	15	2378	66	17	531	36,498	59	20	346

Note: Individuals who did not fit any of these self-reported race/ethnicity categories were excluded from this table.

Abbreviations: LCD, lung cancer deaths; LYG, life-years gained; LYFS-CT, Life-Years From Screening-CT; NNS, number needed to screen; USPSTF, US Preventive Services Task Force.

[a]Individuals who would gain ≥16.2 or ≥12 days of life by the LYFS-CTmodel.

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to determine screening eligibility.[10,37–39] Determining eligibility by benefit from attending screening would make screening more efficient and identify who to prioritize if there was limited capacity. Current guidelines from the American College of Chest Physicians recommend using LYFS-CT to identify individuals with the highest estimated gain in life expectancy from screening who ever-smoked and are ineligible by USPSTF recommendations.[11] Augmenting USPSTF recommendations with such individuals who ever-smoked but are currently ineligible is the most promising approach for improving the fairness, effectiveness, and efficiency, of eligibility for lung screening in the United States.

AUTHOR CONTRIBUTIONS

Rebecca Landy: Conceptualization, formal analysis, and writing – original draft. **Li C. Cheung:** Conceptualization, software, and writing – review and editing. **Corey D. Young:** Writing – review & editing. **Anil K. Chaturvedi:** Conceptualization and writing – review and editing. **Hormuzd A. Katki:** Conceptualization, formal analysis, and writing – original draft.

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CONFLICTS OF INTEREST STATEMENT

The Lung Cancer Risk Assessment Tool (LCRAT), Lung Cancer Death Risk Assessment Tool (LCDRAT) and Life Years From Screening-CT (LYFS-CT) were previously proposed by coauthors of this manuscript.

DATA AVAILABILITY STATEMENT

NHIS data are available from https://www.cdc.gov/nchs/nhis/index.htm. PLCO and NLST data are available for request from https://cdas.cancer.gov/.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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